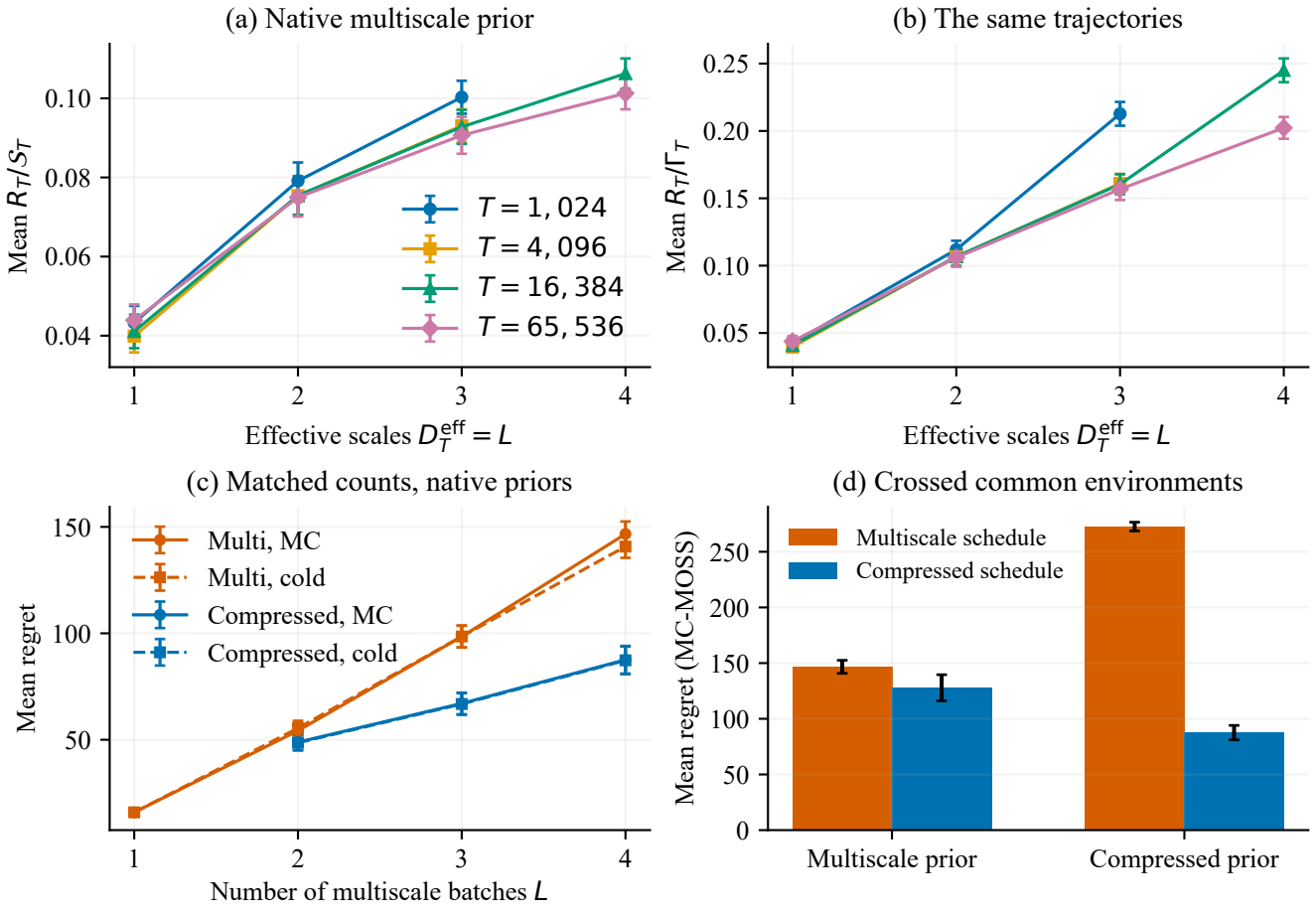
(a) Native multiscale prior
Mean $R_T/S_T$
0.10
0.08
0.06
0.04
$T = 1,024$
$T = 4,096$
$T = 16,384$
$T = 65,536$
1
2
3
4
Effective scales $D_T^{\text{eff}} = L$
(b) The same trajectories
Mean $R_T/\Gamma_T$
0.25
0.20
0.15
0.10
0.05
1
2
3
4
Effective scales $D_T^{\text{eff}} = L$
(c) Matched counts, native priors
Mean regret
150
100
50
Multi, MC
Multi, cold
Compressed, MC
Compressed, cold
1
2
3
4
Number of multiscale batches $L$
(d) Crossed common environments
Mean regret (MC-MOSS)
250
200
150
100
50
0
Multiscale schedule
Compressed schedule
Multiscale prior
Compressed prior